

Numerical Linear Algebra First-Year Exam, January 2024

Do 4 (four) problems.

1. Assume $A \in \mathbb{C}^{m \times m}$.

- (a) Show that A has a Schur decomposition.
- (b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.

2. Let matrix $A \in \mathbb{C}^{m \times n}$ with $n < m$. Let $b \in \mathbb{C}^m$, and let r denote the residual vector $r = b - Ax$.

- (a) Show that x solves the least squares problem $\min \|b - Ax\|_2$ if and only if $r \in \text{Null}(A^*)$.
- (b) Suppose A is full rank, and describe how to find the least squares solution using the QR decomposition of A .

3. Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$ and $\text{rank}(A) = p = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A .

- (a) Using the modified Gram–Schmidt process, write out expressions for q_1, q_2, q_3 , the first three columns of Q in the QR decomposition of A .
- (b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2 .

4. Let $\|\cdot\|$ be a subordinate (induced) matrix norm.

- (a) If E is a $n \times n$ with $\|E\| < 1$, then show $I + E$ is nonsingular and

$$\|(I + E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

- (b) If A is a $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\|\|E\| < 1$, then show $A + E$ is nonsingular and

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

5. If $q_1, \dots, q_n$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$ with $m > n$, prove that the orthogonal projector onto V is QQ^* , where Q is the matrix whose columns are the q_j .